

Exam policy:

- *The exam is 50 minutes (9:00–9:50 AM) and is closed book, closed notes.*
- *You are allowed one hand-written cheat sheet. You can use both sides of one page. Please hand in your cheat sheet with your solutions.*
- *Justify all your answers. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.*
- *No calculators.*

Each of the three questions is worth an equal number of points.

The exam questions are on the other side of this page.

Please do not turn over this page until you are told to start the exam.

Good luck!

Q. 1 (Why did the chicken cross the road?). A chicken wishes to cross a busy road. Cars pass by at the chicken's location according to a Poisson process with an average rate of 3 cars per 10 seconds. The chicken can't keep up with all this fast motion, so she closes her eyes, waits an exponentially distributed time with mean of 10 seconds (independently of the cars), and then crosses the road.

- a. What is the probability that exactly 3 cars pass before the chicken crosses the road?
- b. What is the expected time at which the chicken crosses the road, given that exactly 5 cars have passed before she crosses?

Q. 2 (The easter bunny). The easter bunny arrives on campus. His first stop is Fine Hall (location 1), where he immediately lays one colorful (inexplicably, chicken) egg. From there, he can jump south to Jadwin (location 0) and north to Frist (location 2). All sites further north of Frist and further south of Jadwin are large and dangerous looking construction holes. In each step, the bunny independently jumps one step north, one step south, two steps north, or two steps south, each with probability $\frac{1}{4}$. Every time the bunny jumps, he lays one egg at his new location.

- a. What is the probability that the bunny lays exactly 1 egg at each campus location (Fine Hall, Jadwin, and Frist) before he falls into one of the construction holes?
- b. What is the expected number of jumps the bunny makes until he falls into one of the construction holes?

Q. 3 (Speculation). The price of a stock after t days is modelled as $S_t = e^{B_t}$, where B_t is standard Brownian motion.

- a. What is the probability that the stock price at the end of the second day is exactly twice the stock price at the end of the first day?
- b. What is the probability that the stock price at the end of the first day is at least double where it started (at time 0), but the stock price at the end of the second day has come back down below where it started (at time 0)? Express your answer in terms of the function $\Phi(x) := \int_{-\infty}^x e^{-y^2/2} dy$ (this integral cannot be simplified).

Hint: it may help at some point in your computation to use $2\Phi(x)e^{-x^2/2} = \frac{d}{dx}(\Phi(x)^2)$.